

# Hamromaster

## COMPLETE NOTES

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**For Hamromaster Online  
Classes**

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# Data

\* Statistical data:- Any facts which is expressed numerically

## Types of data:-

1) Primary data: Any information which is calculated by investigator or by his agent for the first time for investigation.

2) Secondary data: Any data which is collected and the investigator collect from that organization is called secondary data.

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## \* Diagrammatic Representation:

- i) Simple bar diagram
- ii) Multiple bar diagram
- iii) Sub-divided bar "
- iv) Pie-chart

## Q. Long question:

\* Making frequency distribution, diagrammatic representation and calculate mean, median or mode from given data.

## \* Frequency distribution:

Simply a table in which the collected and classified data are presented into different classes and the number of cases which fall in each classes are recorded.

## \* Frequency:

The number of times which a variate value occur is known as frequency.

For eg:- The value 5 is repeated 3 times then frequency of 5 is 3.

### \* Types of series :

\* Individual Series :- It is a series where items are used listed simply after observation as distinguished listing them in groups

For eg:- 4, 6, 3, 5, 7, 9

\* Discrete series :- The series formed from a discrete variable is known as discrete series. Also known as frequency distribution.

For eg:- No. of goal score in a match

| No. of goal score in a match | 0  | 1 | 2 | 3 | 4 |
|------------------------------|----|---|---|---|---|
| No. of match                 | 27 | 9 | 3 | 5 | 2 |

\* Continuous series :- The series formed from a continuous variable. Also known as a continuous frequency distribution.

For eg:-

| Wage (in Rs)  | 30-40 | 40-50 | 50-60 | 60-70 | 70-80 |
|---------------|-------|-------|-------|-------|-------|
| No. of worker | 10    | 12    | 15    | 7     | 6     |

### \* Methods of forming class-interval

1) Exclusive method :- The formation of class interval by the method is that upper limit of one class is the lower limit of next class, so as to make continuous without any gap.

0-10 → 1st class (10 = upper limit of 1st class)  
10-20 → 2nd class (10 = lower limit of 2nd class)

For eg:-

| Rainfall (in mm) | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 |
|------------------|-------|-------|-------|-------|-------|
| no. of days      | 12    | 7     | 10    | 4     | 2     |

2) Inclusive method:- The information of class interval by this method is that both lower and the upper limits are included in a particular class.

For eg:-

|                  |       |       |       |       |       |
|------------------|-------|-------|-------|-------|-------|
| Rainfall (in mm) | 10-19 | 20-29 | 30-39 | 40-49 | 50-59 |
| no. of days.     | 12    | 7     | 10    | 4     | 2     |

### Diagrammatic Presentation of Data

Data presented through diagram provides information at a ~~glance~~ glance. They show visual indication of magnitude, grouping, strengths, trends and patterns of the presented data in more attractive and impressive manner.

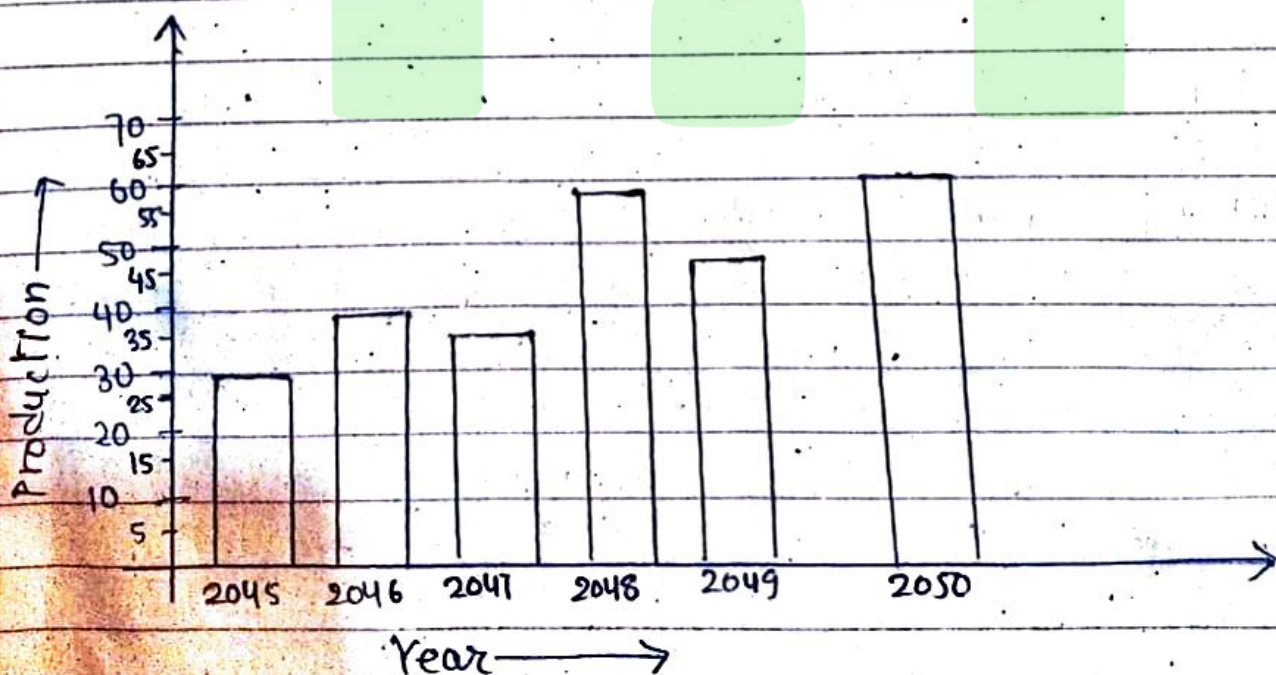
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Types of bar diagrams:-

- 1) Simple b.d
- 2) Sub-divided b.d
- 3) Multiple bar diagram

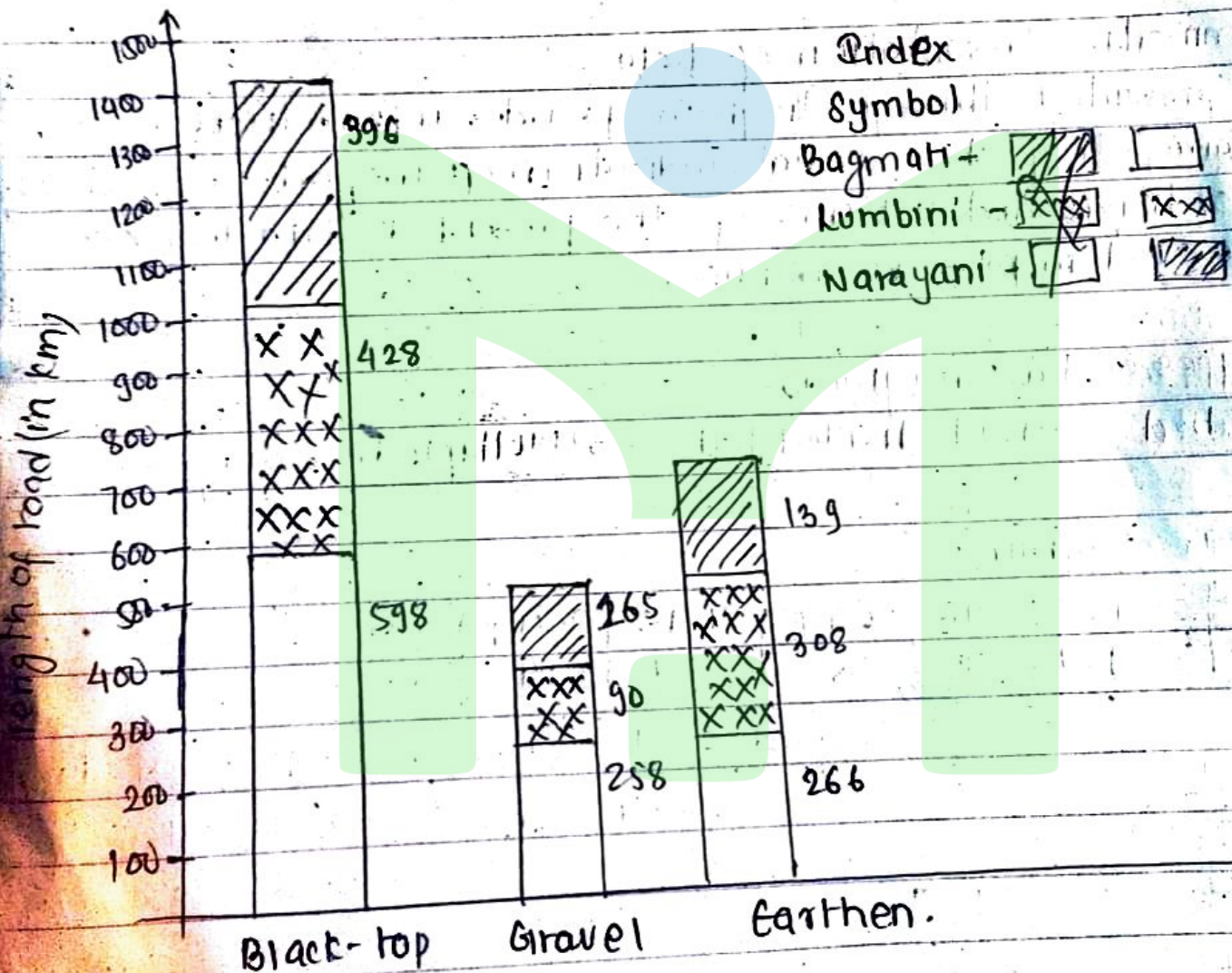
1) Simple bar diagram:-

|            |      |      |      |      |      |      |
|------------|------|------|------|------|------|------|
| Year       | 2045 | 2046 | 2047 | 2048 | 2049 | 2050 |
| Production | 29   | 39   | 35   | 57   | 46   | 60   |



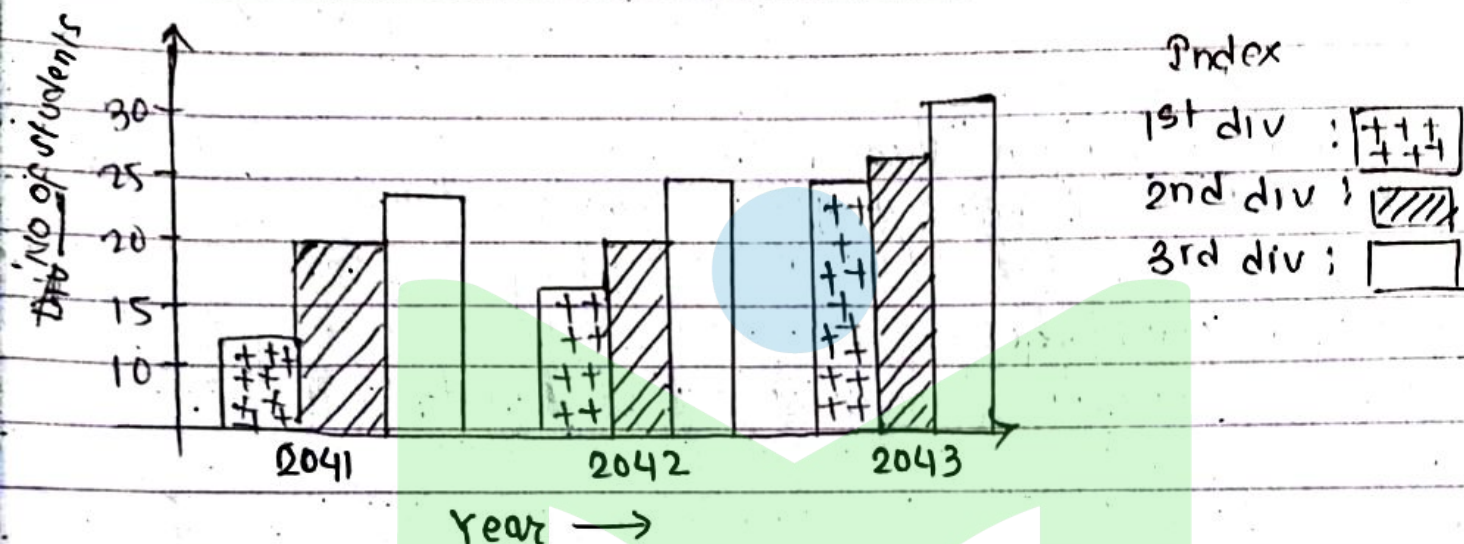
## 2) Sub-divided Bar diagram:

| Zone     | Black-top | Gravel | Earthen | Total-length. |
|----------|-----------|--------|---------|---------------|
| Bagmati  | 598       | 258    | 266     | 1122          |
| Lumbini  | 428       | 90     | 308     | 826           |
| Narayani | 396       | 165    | 139     | 700           |
| Total    | 1422      | 513    | 713     |               |



### 3. Multiple bar diagram:

| Year | 1st div | 2nd div | 3rd div |
|------|---------|---------|---------|
| 2041 | 12      | 15      | 19      |
| 2042 | 17      | 20      | 25      |
| 2043 | 20      | 22      | 26      |



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Pie-charts:

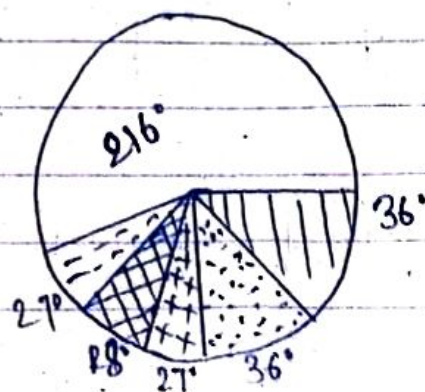
| Items         | Expenditure | Angle                                | Index |
|---------------|-------------|--------------------------------------|-------|
| Food          | 1200        | $1200 \times 0.18^\circ = 216^\circ$ |       |
| Cloth         | 200         | $200 \times 0.18^\circ = 36^\circ$   |       |
| Rent          | 200         | $200 \times 0.18^\circ = 36^\circ$   |       |
| Education     | 150         | $150 \times 0.18^\circ = 27^\circ$   |       |
| Fuel          | 150         | $150 \times 0.18^\circ = 27^\circ$   |       |
| Miscellaneous | 100         | $100 \times 0.18^\circ = 18^\circ$   |       |
| Total         | 2000        | $360^\circ$                          |       |

Soln

Total expenditure = 2000

$$2000 = 360^\circ$$

$$1 = 360 / 2000 = 0.18^\circ$$



## Graphs of frequency distribution

- a) Histogram
- b) Frequency polygon
- c) Frequency
- d) Ogive or cumulative frequency curve.

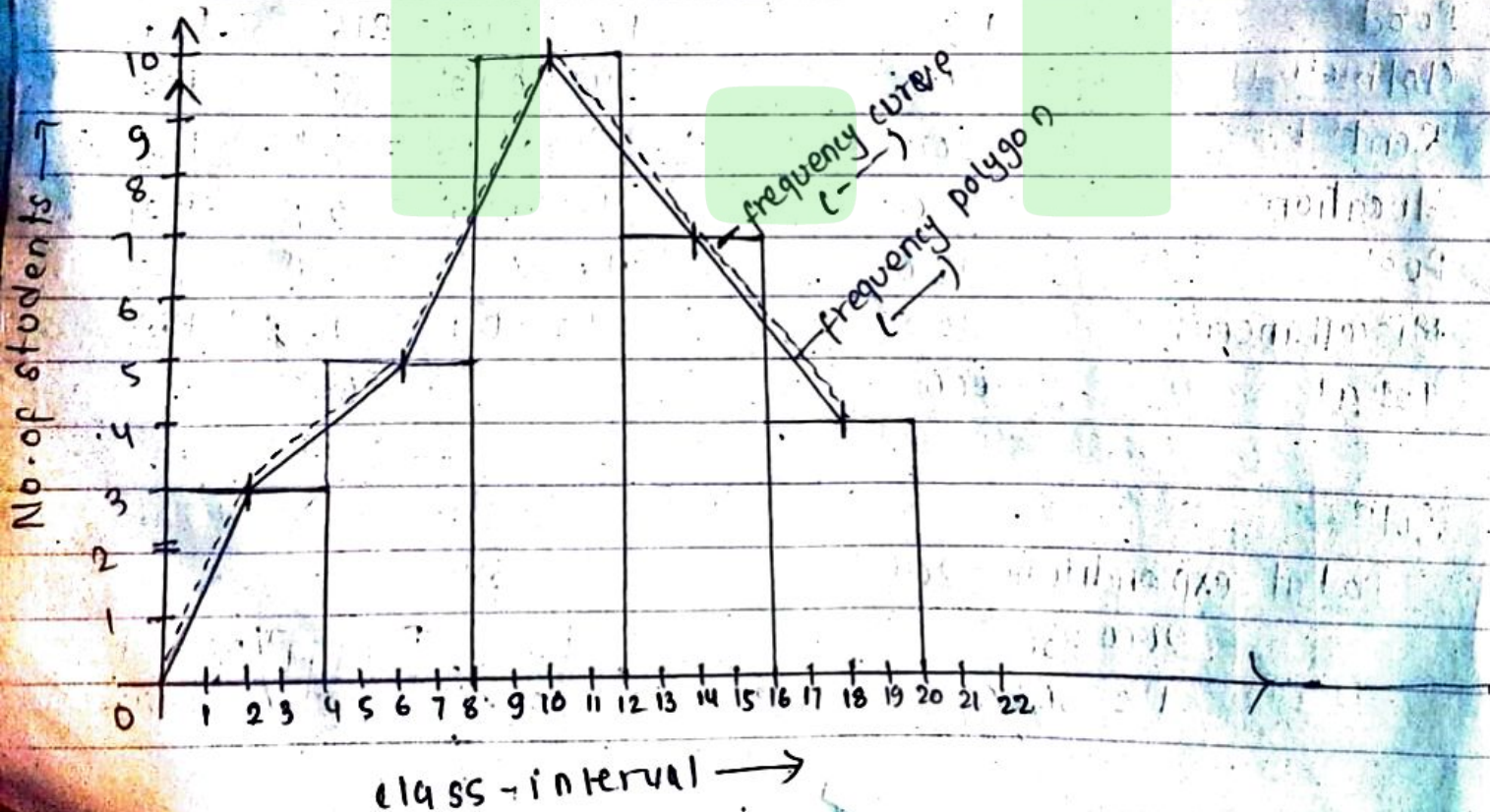
1) Histogram:- Presentation of frequency distribution in the graphical form.

### \* Types of Histogram

- a) Histogram with equal class-size
- b) " " " unequal " "

a) With equal class-size:

| marks | No. of students |
|-------|-----------------|
| 0-4   | 3               |
| 4-8   | 5               |
| 8-12  | 10              |
| 12-16 | 7               |
| 16-20 | 4               |

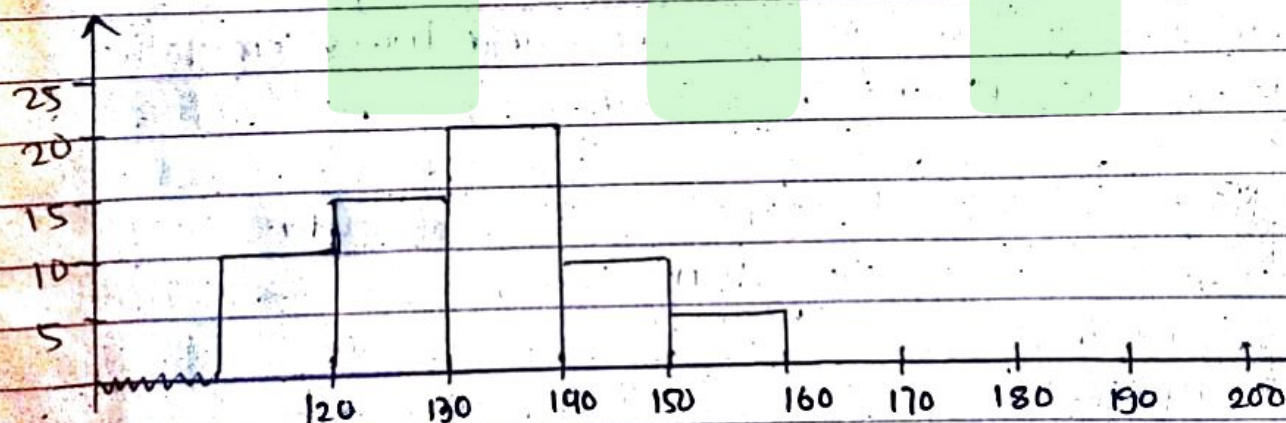


\* For frequency polygon: Join the each mid points of bar diagram by using scale.

\* For frequency curve: Join the each midpoints of bar diagram by using free hand.

b) with unequal class size:

| height (in cm) | No. of person | class-size         | Adjusted frequency |
|----------------|---------------|--------------------|--------------------|
| 120-130        | 10            | 10                 | 10                 |
| 130-140        | 14            | 10                 | 14                 |
| 140-150        | 20            | 10                 | 20                 |
| 150-170        | 16            | $20 = 10 \times 2$ | $16 \div 2 = 8$    |
| 170-200        | 12            | $30 = 10 \times 3$ | $12 \div 3 = 4$    |



## Graphs of frequency distribution

- a) Histogram      b) Frequency polygon  
c) Frequency      d) Ogive or cumulative frequency curve.

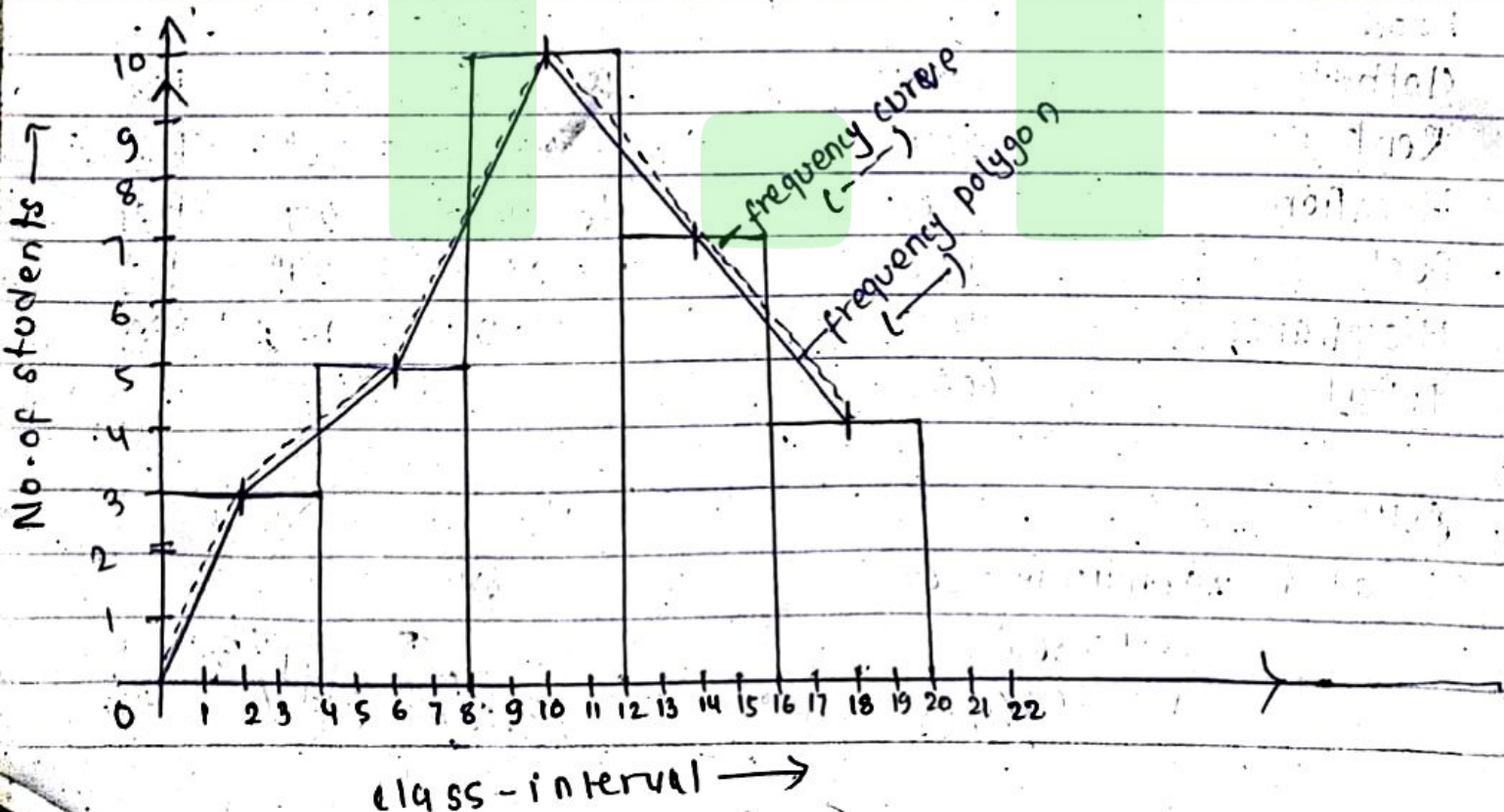
1) Histogram:- Presentation of frequency distribution in the graphical form.

### \* Types of Histogram

- a) Histogram with equal class-size  
b) " " " unequal "

a) With equal class-size:

| marks | No. of students |
|-------|-----------------|
| 0-4   | 3               |
| 4-8   | 5               |
| 8-12  | 10              |
| 12-16 | 7               |
| 16-20 | 3               |

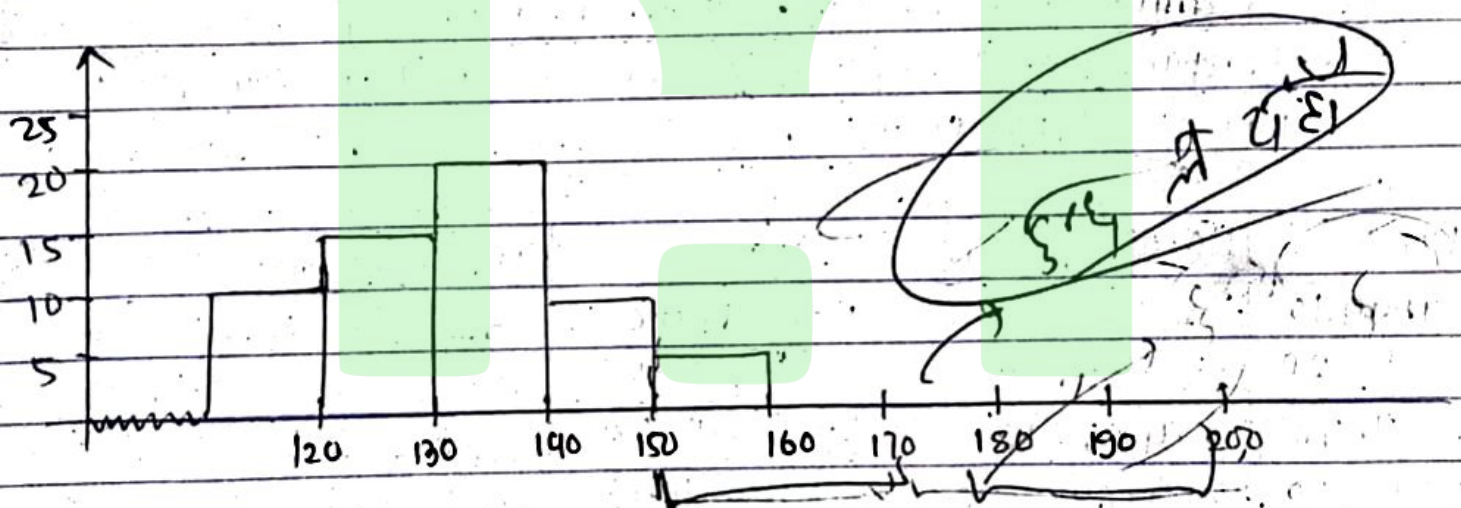


\* For frequency polygon: Join the each mid points of bar diagram by using scale.

\* For frequency curve: Join the each midpoints of bar diagram by using free hand.

b) With unequal class size:

| height (in cm) | No. of person | class-size         | Adjusted frequency |
|----------------|---------------|--------------------|--------------------|
| 120-130        | 10            | 10                 | 10                 |
| 130-140        | 10            | 10                 | 14                 |
| 140-150        | 20            | 10                 | 20                 |
| 150-170        | 16            | $20 = 10 \times 2$ | $16 \div 2 = 8$    |
| 170-200        | 12            | $30 = 10 \times 3$ | $12 \div 3 = 4$    |



Cumulative frequency curve (or ogive)

This curve which is also known as ogive is a graphical representation of cumulative frequency distribution of continuous variable.

In drawing ogive points are plotted with cumulative frequency along y-axis and the corresponding class boundaries along x-axis & joining them freely.

## Types of ogive:

1) Less than ogive:- First prepare less than cumulative frequency in which less the frequencies are added serially from top to bottom. Then, taking upper limits of the class-interval.

2) More than ogive:- First prepare more than cumulative frequencies in which case frequencies are added serially from bottom to the top. Then, taking lower limit of the class-interval.

### Example:

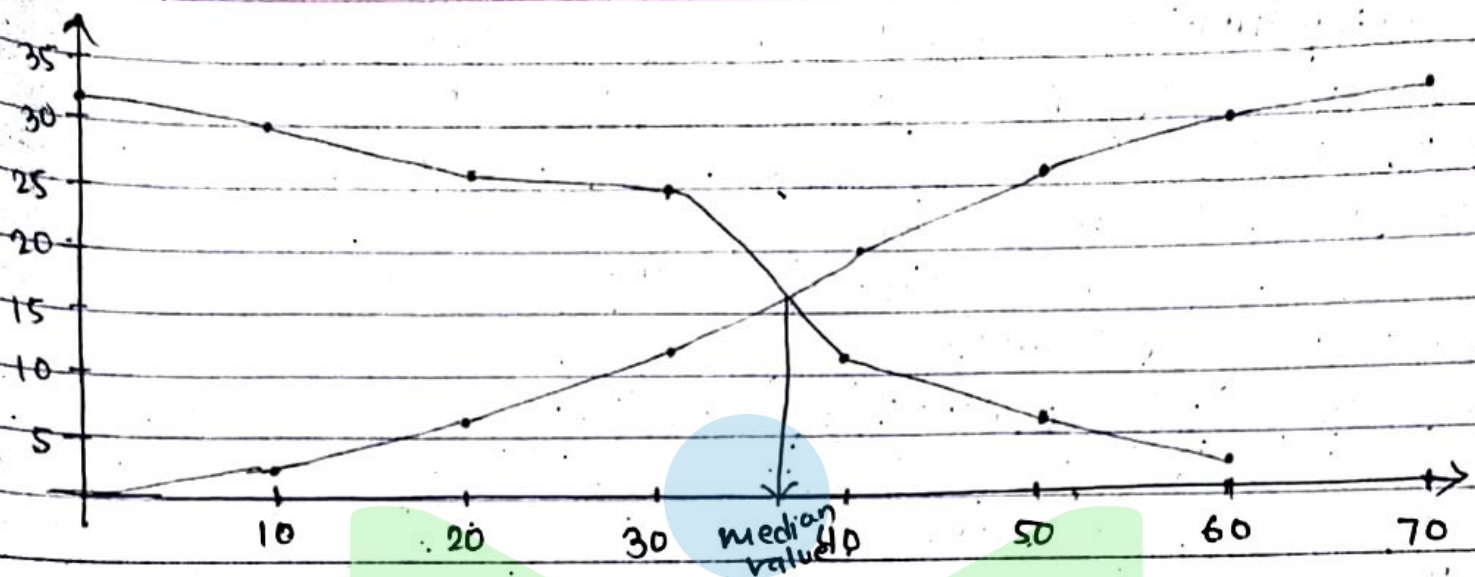
|                 |      |       |       |       |       |       |       |
|-----------------|------|-------|-------|-------|-------|-------|-------|
| Mark            | 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 |
| No. of students | 2    | 4     | 6     | 8     | 6     | 4     | 2     |

Soln

Here, we draw less than as well as more than ogive.

1st we prepare less than ogive & then more than cumulative frequencies distribution.

| Mark         | No. of student     | Mark        | No. of student |
|--------------|--------------------|-------------|----------------|
| less than 10 | 2                  | More than 0 | 32             |
| " " 20       | $2+4=6$            | " " 10      | $32-2=30$      |
| " " 30       | $2+4+6=12$         | " " 20      | $30-4=26$      |
| " " 40       | $2+4+6+8=20$       | " " 30      | $26-6=20$      |
| " " 50       | $2+4+6+8+6=26$     | " " 40      | $20-8=12$      |
| " " 60       | $2+4+6+8+6+4=30$   | " " 50      | $12-6=6$       |
| " " 70       | $2+4+6+8+6+4+2=32$ | " " 60      | $6-4=2$        |



### Measure of central tendency:

- 1) Arithmetic mean
- 2) Median
- 3) Mode
- 4) Geometric mean
- 5) Harmonic mean

### 1) Arithmetic mean: (i) Arithmetic Mean

a) Individual series: 'k'

Let  $x_1, x_2, \dots, x_n$  be the values of the variable then their arithmetic mean is denoted by  $\bar{x}$

Formula:-

(i) Direct method:

$$\bar{x} = \frac{\sum x}{n}$$

where,  $\sum x$  = sum of the observations  
 $n$  = total number of "

(ii) Short-cut method:

$$\bar{x} = A + \frac{\sum d}{n}$$

where,  $A$  = Assumed mean  
 $d = x - A$

Eg:-

The expenditure of 6 families in Rs. are given below:

| Family      | A   | B   | C   | D   | E   | F   |
|-------------|-----|-----|-----|-----|-----|-----|
| Expenditure | 300 | 325 | 375 | 400 | 425 | 500 |

Calculate A.M by (a) Direct  
b) Short-cut Method.

Soln// Using direct method:

$$a) \bar{X} = \frac{\sum x}{n} = \frac{300 + 325 + 375 + 400 + 425 + 500}{6}$$

$$= 387.5$$

Using short-cut method;

Family expenditure

$$d = x - A(375)$$

$$A \quad 300 \quad 300 - 375 = -75$$

$$B \quad 325 \quad -50$$

$$C \quad 375 \quad 0$$

$$D \quad 400 \quad 25$$

$$E \quad 425 \quad 50$$

$$F \quad 500 \quad 125$$

$$\underline{75}$$

$$\bar{X} = A + \frac{\sum d}{n} = 375 + \frac{75}{6} = 387.5$$

b) Discrete series:

Let  $f_1, f_2, \dots, f_n$  be the frequencies of the variate value  $x_1, x_2, \dots, x_n$  resp. Then their arithmetic mean is determined by  $\bar{X}$ .

Formula: Direct method

$$\bar{X} = \frac{\sum fx}{N = (\sum f)}$$

Short-cut method:  $\bar{x} = A + \frac{\sum fd}{n(\sum f)}$

Eg:- (15), Pg-117

Following are the measurements of speed of light in a series of an experiment. Find mean:

12, 30, 30, 27, 30, 39, 18, 27, 48, 24, 18, 25

| $x$ | $f$ | $fx$ | $d = x - A(25)$ | $fxd$ |
|-----|-----|------|-----------------|-------|
| 12  | 1   | 12   | -13             | -13   |
| 18  | 2   | 36   | -7              | -14   |
| 24  | 1   | 24   | -1              | -1    |
| 25  | 1   | 25   | 0               | 0     |
| 27  | 2   | 54   | 2               | 4     |
| 30  | 3   | 90   | 5               | 15    |
| 39  | 1   | 39   | 14              | 14    |
| 48  | 1   | 48   | 23              | 23    |

$N = 12$

$\sum fx = 328$

$\sum fd = 28$

Direct method

$$\bar{x} = \frac{\sum fx}{N}$$

$$= \frac{328}{12}$$

$$= 27.33$$

Short-cut method:

$$\bar{x} = \frac{\sum fd}{N} + A$$

$$= \frac{28}{12} + 25$$

$$= 25.27.33$$

## c) Continuous Series:

Formula: Direct method:  $\bar{X} = \frac{\sum fm}{N}$  where,  $f$  = frequency of class interval  
 $m$  = mid value  
 $= \frac{\text{lower limit} + \text{upper limit}}{2}$

Step deviation method:  $\bar{X} = A + \frac{\sum fd'h}{N}$

where,  $A$  = assumed mean

$d = m - A$  where  $m$  = mid value

$h$  = length of interval

$N$  = total no. of frequency

$d' = \frac{d}{h}$

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Eg: Find A.M. from given data. value:

| Class-interval | frequency | mid-value | $fxm$            | $d = x - A$ | $d' = d/10$ | $fd'$        |
|----------------|-----------|-----------|------------------|-------------|-------------|--------------|
| 15-25          | 4         | 20        | 80               | -20         | -2          | -8           |
| 25-35          | 11        | 30        | 330              | -10         | -1          | -11          |
| 35-45          | 19        | 40        | 760              | 0           | 0           | 0            |
| 45-55          | 14        | 50        | 700              | 10          | 1           | 14           |
| 55-65          | 0         | 60        | 0                | 20          | 2           | 0            |
| 65-75          | 2         | 70        | 140              | 30          | 3           | 6            |
|                | $N = 50$  |           | $\sum fm = 2010$ |             |             | $\sum fd' =$ |

a) Direct method:

$$\bar{X} = \frac{\sum fm}{N} = \frac{2010}{50}$$

$$= 40.2$$

b) Step-deviation method:

$$\bar{X} = A + \frac{\sum fd'h}{N}$$

$$= 40 + \frac{1}{50} \times 10$$

$$= 40.2$$

## (2) Calculation of Median.

The median divides the whole observation into two equal halves one half comparing the values greater and the second half all values less than median. It is denoted by 'Med' or 'Md'.

### 1) Individual series.

a) 1<sup>st</sup> arrange the given data in ascending order or descending order.

b) Use formula  $\text{med} = \left(\frac{N+1}{2}\right)^{\text{th}} \text{ item}$ .

Eg: Find median.

5, 3, 2, 4, 10, 7, 12.

→ Arranging the given data in ascending order;

2, 3, 4, 5, 7, 10, 12

Here,  $n = 7$

$$\text{Med} = \frac{N+1}{2} = \frac{7+1}{2} = 4^{\text{th}} \text{ item} \quad \text{i.e., Med} = 5 \quad \#$$

2. 178, 100, 94, 183, 200, 96, 182, 98, 79, 20

→ Arranging in ascending order;

20, 79, 83, 94, 96, 98, 100, 178, 182, 200

$\therefore n = 10$

$$\text{Med} = \frac{N+1}{2} = \frac{10+1}{2} = 5.5^{\text{th}} \text{ item}$$

$$5.5^{\text{th}} \text{ item} = \frac{5^{\text{th}} \text{ item} + 6^{\text{th}} \text{ item}}{2}$$

$$= \frac{96 + 98}{2}$$

$$= 97 \quad \#$$

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Find median from the given data:

| Marks | No. of students (f) | c.f |
|-------|---------------------|-----|
| 20-30 | 10                  | 10  |
| 30-40 | 20                  | 30  |
| 40-50 | 10                  | 40  |
| 50-60 | 25                  | 65  |
| 60-70 | 10                  | 75  |
| 70-80 | 50                  | 125 |

$$\text{Med} = \frac{N}{2}^{\text{th}} \text{ item} = \frac{125}{2}^{\text{th}} \text{ item}$$

$$= 62.5^{\text{th}} \text{ item}$$

$\therefore 62.5^{\text{th}}$  item lies in class interval of 50-60.

Now,

$$l = 50$$

$$N/2 = 62.5$$

$$h = 10$$

$$c.f = 40$$

$$f = 25$$

Then,

$$\text{Median} = l + \frac{N/2 - c.f}{f} \times h$$

$$= 50 + \frac{62.5 - 40}{25} \times 10$$

$$= 50 + \frac{22.5}{25} \times 10$$

$$= 50 + 9 = 59$$

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### Calculation of mode:

Mode:- Mode is that variate value which repeat maximum number of times, it is the variate value which occurs most frequently in the set of observations.

1) By inspection method:-

For individual series and discrete series.

Eg. Individual series: 2, 3, 4, 5, 5, 6, 7, 8

Here, 5 is repeated 2 times.

$$\therefore \text{Mode} = 5$$

Find median from the given data:

| Marks | No. of students (f) | c.f |
|-------|---------------------|-----|
| 20-30 | 10                  | 10  |
| 30-40 | 20                  | 30  |
| 40-50 | 10                  | 40  |
| 50-60 | 25                  | 65  |
| 60-70 | 10                  | 75  |
| 70-80 | 50                  | 125 |

$$N = 125$$

Then,

$$\begin{aligned} \text{Median} &= l + \frac{N/2 - c.f}{f} \times h \\ &= 50 + \frac{62.5 - 40}{25} \times 10 \\ &= 50 + \frac{22.5}{25} \times 10 \\ &= 50 + 9 = 59 \end{aligned}$$

$$\text{Med} = \frac{N}{2}^{\text{th}} \text{ item} = \frac{125}{2}^{\text{th}} \text{ item}$$

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1) By inspection method:-

For individual series and discrete series.

Eg. Individual series: 2, 3, 4, 5, 5, 6, 7, 8

Here, 5 is repeated 2 times.

$$\therefore \text{Mode} = 5$$

B) Continuous series:-

$$\text{Mode} = j + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

where,  $j$  = lower limit of modal class

$$\Delta_1 = f_1 - f_0$$

$$\Delta_2 = f_1 - f_2$$

$f_0$  = frequency of preceding modal class

$f_1$  = frequency of modal class

$f_2$  = " " following modal class

$h$  = class size.

Eg: Calculate the mode for the distribution of weight of 150 students from the given data below:

| (in kg) weight | Frequency (f) |
|----------------|---------------|
| 30-40          | 18            |
| 40-50          | 37 - $f_0$    |
| 50-60          | 45 - $f_1$    |
| 60-70          | 33 - $f_2$    |
| 70-80          | 15            |
| 80-90          | 8             |

Soln

Since the given distribution is regular and maximum frequency is 45.

Hence the modal class is

50-60

we have, **palpali bharat**

$$j = 50$$

$$\Delta_1 = f_1 - f_0 = 45 - 37 = 8$$

$$\Delta_2 = f_1 - f_2 = 45 - 33 = 12$$

$$h = 10$$

$$\text{Mode} = 50 + \frac{8}{8+12} \times 10$$

$$= 50 + \frac{8}{20} \times 10$$

$$= 54$$

Calculation of mode using Empirical relation:

A distribution with only one mode is known as unimodal. If a distribution have two or more maximum frequency then distribution is known as bimodal or multimodal distribution. In this situation mode is said to be ill-defined and can be calculated by using formula:

$$\text{Mode} = 3 \text{ median} - 2 \times \text{Mean}$$

Partitioned values:

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Quartiles, Deciles, Percentiles.

1) Quartiles: It divides the whole series into 4 equal parts.

$Q_1$  = lower quartile or 1<sup>st</sup> quartiles.

$Q_2$  = 2<sup>nd</sup> quartile or median value.

$Q_3$  = Upper quartile or 3<sup>rd</sup> quartile.

Formula:

|       | Individual series                                     | Discrete series                                          | Continuous series                                                                                                                      |
|-------|-------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $Q_1$ | $\left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$ | $\left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$    | $Q_1 = \frac{N}{4}^{\text{th}} \text{ item}$<br>Class<br>$Q_1 = 1 + \frac{N}{4} - c.f$<br>$\frac{F}{f} \times h$                       |
| $Q_3$ | $\frac{3(N+1)}{4}^{\text{th}} \text{ item.}$          | $3 \left(\frac{N+1}{4}\right)^{\text{th}} \text{ item.}$ | $Q_3 = \frac{3N}{4}^{\text{th}} \text{ item}$<br>$Q_3$ lies in given class<br>$Q_3 = 1 + \frac{3N}{4} - c.f$<br>$\frac{F}{f} \times h$ |

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Formula:

|       | Individual series                                     | Discrete series                                          | Continuous series                                                                                                                              |
|-------|-------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $Q_1$ | $\left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$ | $\left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$    | $Q_1 = \frac{N}{4}^{\text{th}} \text{ item}$<br>class<br>$Q_1 = 1 + \frac{N}{4} - c.f$<br>$\frac{F}{f} \times h$                               |
| $Q_3$ | $\frac{3(N+1)}{4}^{\text{th}} \text{ item.}$          | $3 \left(\frac{N+1}{4}\right)^{\text{th}} \text{ item.}$ | $Q_3 = \frac{3N}{4}^{\text{th}} \text{ item}$<br>$Q_3 \text{ lies in given class}$<br>$Q_3 = 1 + \frac{3N}{4} - c.f$<br>$\frac{F}{f} \times h$ |

## Calculation of Quartile in continuous series:

Eg: 24 Pg no 119

Find the suitable average for the following distribution.

| Weight (in gm) | frequency (f) | c.f | exclusive class: |
|----------------|---------------|-----|------------------|
| 100 or less    | 3             | 3   | 99.5 or less     |
| 101 - 110      | 14            | 17  | 100.5 - 110.5    |
| 110 - 120      | 22            | 39  | 110.5 - 120.5    |
| 120 - 130      | 19            | 58  | 120.5 - 130.5    |
| 131 or more    | 7             | 65  | 130.5 or more.   |
|                | N = 65        |     |                  |

Soln//

Here, the given distribution is open-end class so median is appropriate measure of central tendency.

The given distribution is inclusive class-interval.

Hence, it changed exclusive class-interval.

correction factor (c.f) =  $\frac{\text{lower limit of 2nd class} - \text{upper limit of 1st class}}{2}$

$$= \frac{111 - 110}{2} = \frac{1}{2} = 0.5$$

$$\text{Median} = \frac{N}{2}^{\text{th item}} = \frac{65}{2} = 32.5^{\text{th item}}$$

32.5<sup>th</sup> item is just greater than 39. Hence, the median lies in the 110.5 - 120.5.

$$\begin{aligned} \text{Med} &= l + \frac{\frac{N}{2} - c.f}{f} \times h \\ &= 110.5 + \frac{32.5 - 17}{22} \times 10 \\ &= 117.57 \end{aligned}$$

$$\begin{aligned} Q_1 &= l + \frac{\frac{N}{4} - c.f}{f} \times h \\ &= 100.5 + \frac{16.25 - 3}{14} \times 10 \\ &= 109.36 \end{aligned}$$

$$Q_3 = \frac{3N}{4}^{\text{th}} \text{ item} = 3 \times 16.25 = 48.75^{\text{th}} \text{ item} =$$

class interval : 120.5 - 130.5

$$Q_3 = d + \frac{\frac{3N}{4} - c.f}{f} \times h = 120.5 + \frac{48.75 - 39}{19} \times 10 = 125.63.$$

2) Deciles: It divides whole series into ten equal parts.

Formula:

a) For individual & discrete series

$$D_i = i \left( \frac{N+1}{10} \right)^{\text{th}} \text{ item}$$

$$i = 1, 2, \dots, 9$$

b) For continuous series

$$D_i = i \left( \frac{N}{10} \right)^{\text{th}} \text{ item.}$$

$D_i$  lies in the given class-interval

$$D_i = d + \frac{i \left( \frac{N}{10} \right) - c.f}{f} \times h$$

palpali bharat

3) Percentiles:- It divides the whole observation into 100 equal parts

Formula:

a) For individual & discrete series.

$$P_i = i \left( \frac{N+1}{100} \right)^{\text{th}} \text{ item}$$

$$i = 1, 2, \dots, 99$$

b) For continuous series

$$P_i = i \left( \frac{N}{100} \right)^{\text{th}} \text{ item}$$

$P_i$  lies in the given class-interval

$$P_i = d + \frac{i \left( \frac{N}{100} \right) - c.f}{f} \times h$$

Q. Find  $P_{20}$  in the given problem:

$$P_{20} = \frac{20 \times 65}{100} = 13$$

$P_{20}$  lies in the given class-interval 100.5 - 110.5

$$P_{20} = d + \frac{20 \times \frac{N}{100} - c.f}{f} \times h$$

$$= 100.5 + \frac{13 - 3}{14} \times 10 = 107.64.$$

Combined mean (or mean of composite series)

Let  $\bar{x}_1$  be the mean of  $n_1$  items. Let  $\bar{x}_2$  be the mean of  $n_2$  items.

$\bar{x}$  be the mean of  $n_1 + n_2$  items.

Then, combined mean is given by:

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Eg:- 21 Pg: 118 from text book.

The mean marks got by 300 students in the subject of statistics are 45. The mean of top 100 students was found to be 70 to be 20. What is mean of the remaining 100 students.

Soln

Total number of students,  $N = n_1 + n_2 + n_3 = 100 + 100 + 100 = 300$

Combined mean ( $\bar{x}$ ) = 45

The number of top students ( $n_1$ ) = 100

The mean of top " ( $\bar{x}_1$ ) = 70

The number of last " ( $n_2$ ) = 100

The mean of " " ( $\bar{x}_2$ ) = 20

The number of remaining students ( $n_3$ ) = 100

" mean " " ( $\bar{x}_3$ ) = ?

We have,

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3}{n_1 + n_2 + n_3}$$

$$45 = \frac{100 \times 70 + 100 \times 20 + 100 \times \bar{x}_3}{300}$$

$$\text{or, } 13500 = 9000 + 100 \bar{x}_3$$

$$\text{or, } 100 \bar{x}_3 = 4500$$

$$\text{or, } \bar{x}_3 = 45$$

## Missing frequencies

Eg: 19, Pg 118

Find the missing frequency from the following distribution of sales of shops given that the mean sale of shop is Rs. 24.625.

Soln: Let the missing frequency =  $x$

| Sales | No. of shop | Mid value (m) | $f \times m$ |
|-------|-------------|---------------|--------------|
| 0-10  | 5           | 5             | 25           |
| 10-20 | 25          | 15            | 375          |
| 20-30 | $x$         | 25            | $25x$        |
| 30-40 | 18          | 35            | 630          |
| 40-50 | 7           | 45            | 315          |
|       | $55+x$      |               | $1345+25x$   |

$$\bar{X} = \frac{\sum fm}{N} \Rightarrow 24.625 = \frac{1345+25x}{55+x}$$

$$\text{or, } 24.625 \times 55 + 24.625x = 1345 + 25x$$

$$\text{or, } 0.375x = 9.375$$

$$\text{or, } x = 25$$

## Correct item or correct value:

Eg: 20 Pg: 118

There were 500 workers working in a factory. Their mean was calculated as Rs. 200. Later on it was discovered that the wages of two workers were missed read as Rs. 180 and 20 in place of 80 and 220. Find the correct mean.

Soln:

Total number of worker ( $n$ ) = 500

mean ( $\bar{X}$ ) = 200

Missed read item = 180 & 20

Correct item = 80 & 220

Using formula,  $\bar{x} = \frac{\sum x}{n} \Rightarrow 200 = \frac{\sum x}{500}$

$\sum x = 100000$  from miss read item.

Correct  $\sum x = 100000 - 180 - 20 + 220 + 80$   
 $= 100100$

correct mean =  $\frac{\text{Correct } \sum x}{n} = \frac{100100}{500} = 200.2$

Measure of dispersion:

Dispersion: Dispersion is the degree of the scatter or variation of the variables about a central part.

Measure of dispersion:

1) Range.

3) Mean deviation

2) Quartile deviation (or semi-interquartile range)

4) Standard deviation.

1) Range:- Range may be defined as the difference between the largest value and smallest value.

Formula of range  $(R) = L - S$

where, L = Largest value, S = Smallest value.

coefficient of Range  $= \frac{L - S}{L + S}$

Eg: 30 Pg: 120

The following is the preliminary monthly rainfall data of Kathmandu valley. Calculate range & coefficient of range.

| Months   | Jan | Feb  | Mar | Apr  | May   | Jun   | Jul   | Aug   | Sep   | Oct  |
|----------|-----|------|-----|------|-------|-------|-------|-------|-------|------|
| Rainfall | 6.8 | 15.7 | 8.4 | 34.6 | 179.9 | 250.4 | 498.8 | 460.3 | 145.5 | 20.5 |

| Nov | Dec |
|-----|-----|
| 0.0 | 0.0 |

Largest value (L) = 498.8

Smallest " (S) = 0.0

Range =  $L - S$   
= 498.8

Coefficient of Range =  $\frac{L - S}{L + S}$

$$= \frac{498.8}{498.8} = 1$$

2) Quartile deviation: (or semi-interquartile range)

This method measured the difference between upper quartiles or lower quartiles.

Inter-quartile range =  $Q_3 - Q_1$

Quartile deviation (Q.D) =  $\frac{Q_3 - Q_1}{2}$

Coefficient of quartile deviation =  $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

3) Mean deviation:- (or Average deviation)

Mean deviation is defined as the arithmetic mean of the deviation of the items from mean, median or mode, when all deviation are considered positive.

Formula:-

① Individual Series:

Mean deviation

- from mean =  $\frac{\sum |X - \bar{X}|}{n}$

- from median =  $\frac{\sum |X - \text{Med}|}{n}$

- from mode =  $\frac{\sum |X - \text{mode}|}{n}$

② Discrete & continuous series

Mean deviation

- from mean =  $\frac{\sum f |X - \bar{X}|}{N}$

- from median =  $\frac{\sum f |X - \text{Med}|}{N}$

- from mode =  $\frac{\sum f |X - \text{Mode}|}{N}$

→ coefficient of M.D from

① from mean =  $\frac{\text{M.D from mean}}{\text{mean}}$

② from median =  $\frac{\text{M.D from median}}{\text{median}}$

③ from mode =  $\frac{\text{M.D from mode}}{\text{mode}}$

Eg: 93 Page: 121

12, 6, 7, 3, 15, 10, 18, 5

→ Generally mean deviation from median is calculated if not mentioned

Arranging in ascending order:

Median = size of  $\left(\frac{N+1}{2}\right)^{\text{th}}$  item =  $\left(\frac{8+1}{2}\right)^{\text{th}}$  item = 4.5<sup>th</sup> item.

$$4.5^{\text{th}} = \frac{(4^{\text{th}} + 5^{\text{th}}) \text{ item}}{2} = \frac{7+10}{2} = 8.5$$

| X                              | X - Med (8.5)   |
|--------------------------------|-----------------|
| 3                              | (3 - 8.5) = 5.5 |
| 5                              | 3.5             |
| 6                              | 2.5             |
| 7                              | 1.5             |
| 10                             | 4.5             |
| 12                             | 3.5             |
| 15                             | 6.5             |
| 18                             | 9.5             |
| $\Sigma  x - \text{Med}  = 34$ |                 |

Mean deviation from median:

$$\frac{\Sigma |x - \text{Med}|}{n}$$

$$= \frac{34}{8}$$

$$= 4.25$$

Eg: 34 Page: 121

Calculate mean deviation about mean from the following grouped data.

| Size of item | f  | Mid value | fx              | $ x - \bar{x} $  | $f x - \bar{x} $            |
|--------------|----|-----------|-----------------|------------------|-----------------------------|
| 2-4          | 20 | 3         | 60              | $13 - 5.6 = 2.6$ | $20 \times 2.6 = 52$        |
| 4-6          | 40 | 5         | 200             | 0.6              | $40 \times 0.6 = 24$        |
| 6-8          | 30 | 7         | 210             | 1.4              | $30 \times 1.4 = 42$        |
| 8-10         | 10 | 9         | 90              | 3.4              | $10 \times 3.4 = 34$        |
| $N = 100$    |    |           | $\sum fx = 560$ |                  | $\sum f x - \bar{x}  = 152$ |

$$\bar{x} = \frac{\sum fx}{N} = \frac{560}{100} = 5.6$$

$$\text{Mean deviation from mean} = \frac{\sum f|x - \bar{x}|}{N} = \frac{152}{100} = 1.52$$

Calculate appropriate measure of dispersion.

Soln

The measure of dispersion containing quartile deviation is the proper measure due to open end class.

Here, the given class-interval are not of equal correction for factor

$$C.f = \frac{\text{Lower limit of 2nd class} - \text{Upper limit of 1st class}}{2} = \frac{400 - 399}{2} = 0.5$$

2

2

| Wages         | frequency | Wages            | frequency | C.f. |
|---------------|-----------|------------------|-----------|------|
| less than 200 | 12        | less than 199.5  | 12        | 12   |
| 200-399       | 16        | 199.5 - 399.5    | 16        | 28   |
| 400-599       | 38        | 399.5 - 599.5    | 38        | 66   |
| 600-799       | 78        | 599.5 - 799.5    | 78        | 144  |
| 800-999       | 80        | 799.5 - 999.5    | 80        | 224  |
| 1000-1199     | 35        | 999.5 - 1199.5   | 35        | 259  |
| 1200-1399     | 14        | 1199.5 - 1399.5  | 14        | 273  |
| 1400 & above  | 7         | 1399.5 - & above | 7         | 280  |

Inter quartile range =  $Q_3 - Q_1$

So,

$$Q_3 = 3 \frac{N}{4}^{\text{th}} \text{ item} =$$

$$= 3 \times \frac{280}{4} = 210$$

$$= 799.5 - 999.5$$

$$C.f. = 144$$

$$h = 80$$

$$Q_3 = 1 + \frac{\frac{3N}{4} - C.f.}{f} \times h$$

$$= 799.5 + \frac{210 - 144}{80} \times 80$$

$$= 799.5 + 165$$

$$= 964.5$$

$$Q_1 = \left(\frac{N}{4}\right)^{\text{th}} \text{ item}$$

$$= \frac{280}{4} = 70$$

$$= 599.5 - 799.5$$

$$1 = 599.5 \quad 1 = 200$$

$$C.f. = 66$$

$$f = 78$$

$$Q_1 = 1 + \frac{\frac{N}{4} - C.f.}{f} \times h$$

$$= 599.5 + \frac{70 - 66}{78} \times 200$$

$$= 609.7$$

Coefficient of Quartile deviation =  $\frac{Q_3 - Q_1}{Q_3 + Q_1}$

$$= \frac{964.5 - 609.7}{964.5 + 609.7}$$

$$= \frac{354.8}{1574.2}$$

$$= 0.225$$

$$1574.2$$

$$= 0.225$$

Quartile deviation =

$$\frac{Q_3 - Q_1}{2} = \frac{354.8}{2}$$

$$= 177.4$$

## Standard Deviation

The standard deviation is defined as the positive square root of arithmetic mean of the square deviation from their arithmetic mean of set of values. It is also known as root-mean square deviation. It is usually denoted by the Greek letter  $\sigma$  (small sigma).

Formulae: Individual series.

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

i) Direct method:  $\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$

ii) Short cut method:

$$\sigma = \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2}$$

Where,  $d = x - \text{Assumed mean}$

iii) Actual mean method:  $\sigma = \frac{\sum x^2}{n}$  where  $x = X - \bar{X}$

2) Discrete and continuous series:

Standard deviation:  $\sigma = \sqrt{\frac{\sum f (X - \bar{X})^2}{n}}$

i) Direct method:  $\sigma = \sqrt{\frac{\sum f X^2}{N} - \left(\frac{\sum f X}{N}\right)^2}$

ii) Actual mean method:  $\sigma = \sqrt{\frac{\sum f x^2}{N}}$  where  $x = X - A$

iii) Short-cut method ( $\sigma$ ):  $\sigma = \sqrt{\frac{\sum f d^2}{n} - \left(\frac{\sum f d}{n}\right)^2}$  where,  $d = X - A$   
Assumed value

Example:

Ten measurements were made with the following results length in cm

70, 73, 75, 70, 72, 76, 75, 72, 74, 76

Find standard deviation by direct method & short-cut method.

Direct method

| length (cm) / x          | $X - \bar{X} = 3$ | $(X - \bar{X})^2$             |
|--------------------------|-------------------|-------------------------------|
| 77                       | $77 - 74 = 3$     | 9                             |
| 73                       | -1                | 1                             |
| 75                       | +1                | 1                             |
| 70                       | -4                | 16                            |
| 72                       | -2                | 4                             |
| 76                       | 2                 | 4                             |
| 75                       | 1                 | 1                             |
| 72                       | -2                | 4                             |
| 74                       | 0                 | 0                             |
| 76                       | 2                 | 4                             |
| $\Sigma = 740 / 10 = 74$ |                   | $\Sigma (X - \bar{X})^2 = 44$ |

Short-cut method

| $d = X - A (76)$ | $d^2$ |
|------------------|-------|
| $77 - 76 = 1$    | 1     |
| $73 - 76 = -3$   | 9     |
| -1               | 1     |
| -6               | 36    |
| -4               | 16    |
| 0                | 0     |
| -1               | 1     |
| -4               | 16    |
| -2               | 4     |
| 0                | 0     |
| $\Sigma = 84$    |       |

Direct method

$$\begin{aligned}
 S.D (6) &= \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \\
 &= \sqrt{\frac{44}{10}} \\
 &= \sqrt{4.4} \\
 &= 2.1
 \end{aligned}$$

Short-cut method.

$$\begin{aligned}
 S &= \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} \\
 &= \sqrt{\frac{84}{10} - \left(\frac{-20}{10}\right)^2} \\
 &= \sqrt{8.4 - 4} \\
 &= \sqrt{4.4} = 2.1
 \end{aligned}$$

Example:

Find the mean and standard deviation from the following data.

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| x | 10 | 11 | 12 | 13 | 14 |
| f | 8  | 12 | 18 | 12 | 3  |

| x  | f      | fx  | d = (x - $\bar{x}$ ) (12) | fd              | fd <sup>2</sup> (1d x d = 1d <sup>2</sup> ) |
|----|--------|-----|---------------------------|-----------------|---------------------------------------------|
| 10 | 3      | 30  | 10 - 12 = -2              | -6              | 12                                          |
| 11 | 12     | 132 | 11 - 12 = -1              | -12             | 12                                          |
| 12 | 18     | 216 | 12 - 12 = 0               | 0               | 0                                           |
| 13 | 12     | 156 | 13 - 12 = 1               | 12              | 12                                          |
| 14 | 3      | 42  | 14 - 12 = 2               | 6               | 12                                          |
|    | N = 48 | 516 | $\Sigma$                  | $\Sigma fd = 0$ | $\Sigma fd^2 = 48$                          |

$$\bar{x} = \frac{\sum fx}{N} = \frac{516}{48} = 12$$

$$\begin{aligned}
 S &= \sqrt{\frac{\sum fd^2}{n} - \left(\frac{\sum fd}{n}\right)^2} \\
 &= \sqrt{\frac{48}{48} - \left(\frac{0}{48}\right)^2} \\
 &= \sqrt{1} = 1
 \end{aligned}$$

Direct method

$$\begin{aligned} \text{S.D (6)} &= \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \\ &= \sqrt{\frac{44}{10}} \\ &= \sqrt{4.4} \\ &= 2.1 \end{aligned}$$

Short-cut method.

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum d^2}{n} - \left(\frac{\sum d}{n}\right)^2} \\ &= \sqrt{\frac{84}{10} - \left(\frac{-20}{10}\right)^2} \\ &= \sqrt{8.4 - 4} \\ &= \sqrt{4.4} = 2.1 \end{aligned}$$

Example:

Find the mean and standard deviation from the following data.

|   |    |    |    |    |    |
|---|----|----|----|----|----|
| X | 10 | 11 | 12 | 13 | 14 |
| f | 3  | 12 | 18 | 12 | 3  |

| X  | f      | fx  | d = (x - $\bar{x}$ ) (12) | fd              | fd <sup>2</sup> (1d x d = 1d) |
|----|--------|-----|---------------------------|-----------------|-------------------------------|
| 10 | 3      | 30  | 10 - 12 = -2              | -6              | 12                            |
| 11 | 12     | 132 | 11 - 12 = -1              | -12             | 12                            |
| 12 | 18     | 216 | 12 - 12 = 0               | 0               | 0                             |
| 13 | 12     | 156 | 13 - 12 = 1               | 12              | 12                            |
| 14 | 3      | 42  | 14 - 12 = 2               | 6               | 12                            |
|    | N = 48 | 516 | $\Sigma$                  | $\Sigma fd = 0$ | $\Sigma fd^2 = 48$            |

$$\bar{x} = \frac{\sum fx}{N} = \frac{516}{48} = 12$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum fd^2}{n} - \left(\frac{\sum fd}{n}\right)^2} \\ &= \sqrt{\frac{48}{48} - \left(\frac{0}{48}\right)^2} \\ &= \sqrt{1} = 1 \end{aligned}$$

9-11

Coefficient of variation:

It is a relative measure of dispersion based on standard deviation in order to compare variability between two sets of data.

Coefficient of variation is often abbreviated as C.V. & calculated as

For 
$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

For comparing the variability between two or more than two sets of data. A distribution having more C.V. is considered less consistency, less uniform, or less equitable, or less stable.

A distribution having less C.V. is considered more homogeneous, more consistent, more uniform, more equitable or more stable.

Eg: 37 Pg: 121

The masses of  $n$  grams to the contents of 25 items of brand

A anchovies are summarized by  $\sum x = 1268.2$  &  $\sum x^2 = 64585.16$

Find mean and variance of the masses also find coefficient of variation.

Soln//

$N = 25$

$\sum x = 1268.2$

$\sum x^2 = 64585.16$

Then,

Mean =  $\frac{\sum x}{N}$

$= \frac{1268.2}{25}$

$= 50.728$

Standard deviation;

$$\sigma = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$= \sqrt{\frac{64585.16}{25} - (50.728)^2}$

$= 3.3$

Variance  $(\sigma)^2 = (3.3)^2$

Coefficient of variation;

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

$= \frac{3.3}{50.728} \times 100$

$= 6.505$

$= 0.6\%$

Relation bet<sup>n</sup> standard deviation & range by approximation relation

1. Standard deviation ( $\sigma$ ) =  $\frac{\text{Range}}{4}$

Combined standard deviation =  $\sigma_{12} = \sqrt{\frac{1}{n_1 + n_2} [n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)]}$

where,  $d_1 = \bar{x}_1 - \bar{x}_2$  &  $d_2 = \bar{x}_2 - \bar{x}_{12}$

where,  $\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$

## \* Skewness \*

Skewness is a lack of symmetry. Measure of skewness is a descriptive statistical measure used to measure the shape of the curve drawn from the frequency distribution or to measure the direction of variation.

Types of skewness:

- 1) No skewness (or symmetrical) :- Mean = Median = mode
- 2) Positive " :- Mean > Median > Mode
- 3) Negative " :- Mean < Median < Mode.

\* Bowley's coefficient of skewness:

The absolute measure of skewness is, ~~skewness~~

$$\text{skewness} = \frac{(Q_3 - \text{Med}) + (\text{Med} - Q_1)}{Q_3 + Q_1 - 2\text{Med}}$$

The relative measure of Bowley's coefficient of skewness is Bowley's coefficient of skewness.

$$Sk(B) = \frac{Q_3 + Q_1 - 2Md}{Q_3 - Q_1}$$

This coefficient of skewness is known as quartile coefficient of skewness. Its value lies betn  $-1$  to  $+1$ . i.e.  $-1 \leq SK(B) \leq +1$ .

Interpretation of result;

- 1) If  $SK(B) = 0$ , the distribution is symmetrical.
- 2) If  $SK(B) > 0$ , " " " positively "
- 3) If  $SK(B) < 0$ , " " " negatively "

Eg: (5) pg: 125

| Height in cm | f   | C.f |
|--------------|-----|-----|
| less than 50 | 40  | 40  |
| 50-100       | 80  | 120 |
| 100-150      | 130 | 250 |
| 150-200      | 60  | 310 |
| 200 & above  | 30  | 340 |
| n = 340      |     |     |

Soln: Here, the given problem is of end class. So, we use Bowley's coefficient of skewness.

we have,  

$$SK(B) = \frac{Q_3 + Q_1 - 2Med}{Q_3 - Q_1}$$

For  $Q_1$ ,

$$Q_1 = \frac{N}{4} = \frac{340}{4} = 85^{th} \text{ item}$$

which lies in class 50-100

as 120 is just greater than 85<sup>th</sup> item.

$$Q_1 = L + \frac{N/4 - C.f}{f} \times h$$

$$= 50 + \frac{85 - 40}{80} \times 50$$

$$= 78.125$$

For  $Q_3$ ,

$$Q_3 = \frac{3N}{4} = \frac{3 \times 340}{4} = 255^{th} \text{ item}$$

which lies in class 150-200, as 310 is just greater than 255.

$$Q_3 = L + \frac{3N/4 - C.f}{f} \times h$$

$$= 150 + \frac{255 - 250}{60} \times 50$$

$$= 154.166$$

For Median,

$$N_2 = \frac{340}{2} = 170^{th} \text{ item}$$

class 100-150

$$Med = L + \frac{N_2 - C.f}{f} \times h$$

$$= 100 + \frac{170 - 120}{80} \times 50$$

$$= 119.23$$

Using formula;

$$SK = \frac{Q_3 + Q_1 - 2Med}{Q_3 - Q_1}$$

$$Q_3 - Q_1$$

$$= -0.081$$

Since,  $SK < 0$  means it is negatively skewed.

## \* Karl - Pearson's Measure to skewness:

The absolute measure of skewness are:-

1) skewness = mean - mode

2) skewness = mean - median

The absolute measure of skewness are not widely used because they are expressed in terms of original units of data. The relative measure of skewness is known as coefficients of skewness and is used more frequently.

→ The coefficient of skewness ( $sk$ ) is defined by Karl Pearson is given by:

It lies between  $-1 \leq sk(P) \leq +1$

$$sk(P) = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{\bar{X} - \text{Mode}}{\sigma} \quad \text{--- (i)}$$

$$sk(P) = \frac{3(\text{Mean} - \text{Median})}{\text{Standard deviation}} = \frac{3(\bar{X} - \text{Med})}{\sigma} \quad \text{--- (ii)}$$

It lies between  $-3 \leq sk(P) \leq +3$

## \* Interpretation of result:

- 1) If  $sk(P) = 0$ , the distribution is symmetrical.
- 2) If  $sk(P) > 0$ , the distribution is positively skewed.
- 3) If  $sk(P) < 0$ , the distribution is negatively skewed.

50) Calculate Karl Pearson's coefficient of skewness of the following grouped data & interpret the result.

| Class-Interval | No. of plants (f)    | Mid-value (M) | fm   | fm <sup>2</sup> |
|----------------|----------------------|---------------|------|-----------------|
| 1) 5-10        | 2                    | 7.5           | 15   | 112.5           |
| 2) 10-15       | 5 $\rightarrow f_1$  | 12.5          | 62.5 | 781.25          |
| 3) 15-20       | 10 $\rightarrow f_m$ | 17.5          | 175  | 3062.5          |
| 20-25          | 4 $\rightarrow f_2$  | 22.5          | 90   | 2025            |
| 25-30          | 1                    | 27.5          | 27.5 | 756.25          |
|                | $N = 22$             |               | 370  | 16737.5         |

1) Value 1st increase then maximum & then decrease then  
2) Use mode in formula

$$\text{Mean} = \frac{\sum fm}{N} = \frac{370}{22} = 16.81$$

$$\text{Standard deviation (S)} = \sqrt{\frac{\sum fm^2}{N} - \left(\frac{\sum fm}{N}\right)^2}$$

(Direct method)

$$= \sqrt{\frac{16737.5}{22} - \left(\frac{370}{22}\right)^2}$$

$$= 4.6$$

for Mode 10 (maximum frequency)

Mode lies in the class 15-20.

$$\therefore \text{Mode} = 1 + \frac{f_m - f_1}{2f_m - f_1 - f_2} \times h$$

$$= 15 + \frac{10 - 5}{20 - 5 - 4} \times 20 = 15 + \frac{5 \times 20}{11} = 17.27$$

Moment chapter  $\rightarrow$  Don't have to read.  
raw moment, central moment (not in syllable)

$\therefore$  Karl Pearson's coefficient of skewness ( $Sk$  or  $P$ ) =  $\frac{\text{Mean} - \text{Mode}}{\text{standard deviation}}$

$$= \frac{16.81 - 17.27}{4.86}$$

$$= -0.8 - 0.088 < 0$$

If  $Sk < 0$ , the distribution is negatively skewed.

11

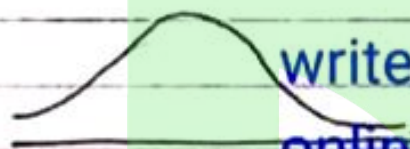
## Measure of kurtosis

Measure of kurtosis is descriptive statistical measure used to measure the flatness or peakedness of the curve drawn from frequency distribution. This is useful to test the normality of the curve.

### Kinds of kurtosis:

#### 1) Mesokurtic curve:

A curve is said to be mesokurtic if the curve drawn from the frequency distribution is neither so flat or not so peaked. This is moderate type of curve or normal curve.

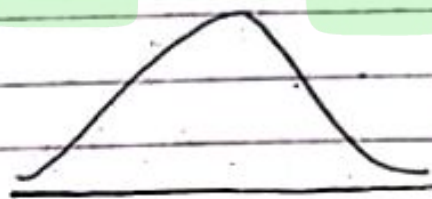


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#### 2) Leptokurtic curve:-

A curve is said to be leptokurtic if the curve drawn from the frequency distribution is more peaked than mesokurtic curve.



#### 3) Platykurtic:



### iii) Platykurtic curve:

A curve drawn from the frequency distribution is very flat than mesokurtic curve.

### Percentile measure of kurtosis:

The measure of kurtosis based on both quartiles and percentile is percentile coefficient of kurtosis and is given by as;

$$\text{Kurtosis (K)} = \frac{1}{2} \frac{(Q_3 - Q_1)}{P_{90} - P_{10}}$$

### XX Interpretation of result:

- 1) If  $K \approx 0.263$ , the distribution is mesokurtic or normal.
- 2) If  $K > 0.263$ , the distribution is leptokurtic.
- 3) If  $K < 0.263$ , the distribution is platykurtic.

### Exercise Q.N 54 Pg. 125

The following table shows the number of children for 35 families in a certain area. Find the percentile coefficient of kurtosis and interpret the value.

| no. of children (x) | no. of families (f) | C.f |
|---------------------|---------------------|-----|
| 0                   | 3                   | 3   |
| 1                   | 5                   | 8   |
| 2                   | 12                  | 20  |
| 3                   | 9                   | 29  |
| 4                   | 4                   | 33  |
| 5                   | 2                   | 35  |

For  $Q_1 = \left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$   
 $= \left(\frac{35+1}{4}\right)^{\text{th}} = 9^{\text{th}} \text{ item}$

The c.f. is just greater than 9 is 20.  
 $\therefore Q_1 = 20$

For  $Q_3 = 3\left(\frac{N+1}{4}\right)^{\text{th}} \text{ item}$   
 $= 3 \times 9 = 27^{\text{th}} \text{ item}$

The c.f. just greater than 27 is 29.  
 $\therefore Q_3 = 29$

For  $P_{10} = 10\left(\frac{N+1}{100}\right) = 10 \times \left(\frac{35+1}{100}\right) = \frac{36}{10} = 3.6^{\text{th}} \text{ item}$

The c.f. is just greater than 3.6 is 8.  
 $\therefore P_{10} = 8$

For  $P_{90} = 90\left(\frac{N+1}{100}\right) = \frac{9 \times 36}{10} = \frac{324}{10} = 32.4^{\text{th}} \text{ item}$

The c.f. just greater than 32.4 is 33.  
 $\therefore P_{90} = 33$

$$K = \frac{\frac{1}{2}(Q_3 - Q_1)}{P_{90} - P_{10}} = \frac{\frac{1}{2}(29 - 20)}{\frac{33 - 8}{4 - 1}} = \frac{1}{2 \times 3} = \frac{1}{6} = 0.166$$

Since,  $K = 0.166 < 0.283$ , the distribution is platykurtic.

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## COMPLETE NOTES

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